

USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #4 (Lesson 17) - Introduction to Phased Array Hardware
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.3: I can identify the hardware architecture of the ADALM-PHASER and control it via SDR software.

This sheet is the turn-in document for the Lesson 17 lab. Part A records how your team brought its kit up from a blank card, and Part B records the first measurements on that kit. Fill in every box, and show the arithmetic where a box asks for a calculated value. The lab is a team assignment, so each team submits one sheet.

Part A: Bring Up Your PHASER

- 1. The Bring-Up Record.** Record what you set and what the kit reported as you brought it up. Copy the installer's lines exactly as it prints them under [6b/6] Checking the RF chain....

Item	Value
Team number NN	
Hostname in <code>phaser-hostname</code>	
Fixed IP in <code>phaser-ip</code>	
Address you opened the UI at	
Wi-Fi join: result of <code>ping -c 4 github.com</code>	
Installer line HB100	
Installer line Rx_freq	
Installer line L0	
Installer line ADF4159	
Frequency Find HB100 reported	

- 2. The Identity Reset.** Every team flashed its card from the same golden image, and on its first boot each card regenerates its SSH host keys and its `/etc/machine-id` before the ssh server starts. Explain in three or four sentences why a cloned card must do this.

Part B: First Measurements

3. **The Frequency Plan.** Copy the frequency **Find HB100** reported in Part A, then work out where the LO must sit for it. The mixer’s output is the LO minus the source, and the IF is a setting, 2.2 GHz, chosen so that the LO’s range covers every HB100; so the LO is the only frequency that moves when the source moves.

Quantity	Value
Measured HB100 frequency f_{RF}	
Required LO frequency f_{LO}	
Resulting IF f_{IF}	
Is the LO inside 12.2–13.0 GHz?	

Show the arithmetic for your measured frequency here.

- 4. The Labeled Block Diagram.** Sketch the receive chain from the patches to the Raspberry Pi. Label every block with its name and with the frequency present at that point, mark where the analog summing happens, and mark where the two digital channels begin.



5. FFT Observations. Work on the FFT tab with the source at 1 m and record what the spectrum does as you change two controls.

(a) Where does the tone sit on the FFT tab's frequency axis, and how far above the noise floor is it?

Peak offset =

Peak above floor =

(b) Set **Rx Gain (dB)** to 10 and record the peak and floor, then repeat at 0 and at 20. Report the peak and floor levels in dBFS (decibels relative to the ADC's full-scale input) and the separation between them in dB.

Rx Gain	Peak (dBFS)	Floor (dBFS)	Separation (dB)
0 dB			
10 dB			
20 dB			

(c) Lower **Signal Freq (GHz)** by 0.001 GHz: click the field's down arrow once, or type the new value and press Enter. Record how far the peak moves and in which direction. (If a typed value does not move the peak, press **Stop** and then **Start**.)

Peak shift =

6. Written Answers. Answer each in three or four sentences.

- (a) Why does the control software have to measure the HB100's frequency instead of assuming 10.525 GHz?

- (b) The array has eight elements, but software sees only two digital channels. Explain where the other six went, and name one measurement this rules out.

Documentation: _____